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Estimation of subsurface temperature anomaly in the Indian Ocean during recent global surface warming hiatus from satellite measurements: A support vector machine approach

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Highlights

- A new approach to estimate subsurface temperature anomaly from remote sensing
- SVM can estimate STA upper 1000m in the Indian Ocean from sea surface features.
- A helpful technique for studying subsurface and deeper ocean thermal variability
- The intensity of STA was negatively related to the depth.

Article preview

Abstract

Introduction

Section snippets

References (32)

Abstract

Estimating the thermal information in the subsurface and deeper ocean from satellite measurements over large basin-wide scale is important but also challenging. This paper proposes a support vector machine (SVM) method to estimate subsurface temperature anomaly (STA) in the Indian Ocean from a suite of satellite remote sensing measurements including sea surface temperature anomaly (SSTA), sea surface height anomaly (SSHA), and sea surface salinity anomaly (SSSA). The SVM estimation of STA features the inclusion of in-situ Argo STA data for training and testing. SVM, one of the most popular machine learning methods, can well estimate the STA in the upper 1000m of the Indian Ocean from satellite measurements of sea surface parameters (SSTA, SSHA and SSSA as input attributes for SVM). The results, based on the common SVM application of Support Vector Regression (SVR), were validated for accuracy and reliability using the Argo STA data. Both MSE and r^2 for performance measures are improved after including SSSA for SVR (MSE decreased by 12% and r^2 increased by 11% on average). The results showed that SSSA, in addition to SSTA and SSHA, is a useful parameter that can help detect and describe the deeper ocean thermal structure, as well as improve the STA estimation accuracy. Moreover, our method can provide a useful technique for studying subsurface and deeper ocean thermal variability which has played an important role in recent global surface warming hiatus since 1998, from satellite measurements in large basin-wide scale.

Keywords

Subsurface temperature anomaly; Satellite measurements; Support vector machine; The Indian Ocean; Global warming hiatus

Introduction

Accurately detecting and describing the subsurface thermal structure of the global oceans are important aspects of ocean dynamics. Subsurface and deeper ocean thermal remote sensing is becoming even more important since new evidence suggests widespread warming in the world's deeper ocean as a response to the global climate variability and change (Balmaseda et al., 2013, Song and Colberg, 2011). Satellite remote sensing provides many useful ocean surface observations at various spatial and temporal scales, but is limited to the surface layers of the ocean (Ali, Swain, & Weller, 2004). Since many important ocean processes and features are located well below the sea surface and at considerable depths (Klemas & Yan, 2014), it is essential to determine the extent to which such surface remote sensing observations can be used to develop information about the ocean's interior (Ali et al., 2004). Subsurface and deeper ocean remote sensing has the ability to derive ocean interior dynamic parameters (especially the thermohaline structure) and enables us to characterize subsurface and deeper ocean processes and features and their implications for the climate change (Klemas & Yan, 2014).

The estimation of the 3-D thermohaline structure over large spatial scales is critical to physical oceanographers, as well as to biological and chemical oceanographers, because they can estimate subsurface flow fields, allowing them to compute horizontal and vertical advection in the ocean interior (Wilson & Coles, 2005). This is becoming even more significant than ever since there are indications that the deeper ocean is warming up and has played an important role in recent global surface warming hiatus since 1998 (Chen and Tung, 2014, Drijfhout et al., 2014).

Since the subsurface thermal information cannot be directly observed from satellites remote sensing, most of the related research has traditionally relied on numerical modeling and in situ measurements that are not able to provide suitably wide coverage in near-real time (Fiedler, 1988). Thus, improvements in the accuracy of large-scale subsurface thermal structure in near-real time assimilation models are needed. As an alternative to obtaining subsurface information from assimilated models, satellite measurements can be used if special algorithms/techniques are developed (Klemas & Yan, 2014).

Many subsurface phenomena have surface manifestations that can be interpreted with the help of satellite measurements and numerical models to derive key parameters of deeper ocean processes. Khedouri, Szczechowski, and Cheney (1983) attempted to estimate subsurface temperature from satellite altimetry. Chu, Fan, and Liu (2000) showed that a parametric model could determine the subsurface thermal structure from satellite SST observations. Fischer (2000) estimated the vertical thermal structure at the equator using both sea surface height anomaly (SSHA) and SST anomaly (SSTA) based on a multivariate projection method. In addition to regression methods, Ali et al. (2004) showed that the ocean subsurface thermal structure could be determined from surface parameters using a neural network.

There are a few other studies of the ocean interior using satellite measurements (Fox et al., 2002, Meijers et al., 2011, Swart et al., 2010, Willis et al., 2003). Recently, Wu, Yan, Jo, and Liu (2012) also used multiple satellite measurements and a neural network algorithm, Self-Organizing Map (SOM), to estimate subsurface temperature profiles. Guinehut et al., 2004, Guinehut et al., 2009, Guinehut et al., 2012, Guinehut et al., 2006 also reported on improved high resolutions of 3-D temperature and salinity fields derived from Argo data and altimetry.

There are high spatial similarities between the STA and dynamic height anomaly (DHA) at different depths (Dennis et al., 2001, Willis et al., 2003). The simplest way to estimate STA from sea surface change information was to compute linear regression coefficients between STA at different depth and DHA based on historic measurements (Willis et al., 2003), and then to multiply the coefficients of DHA to obtain STA at each depth.

In general, the strategies for deriving subsurface thermal information from surface parameters are based on either the combination of dynamical models and in situ observations or purely on statistical relationships between the surface and the subsurface parameters. All of the previous statistical approaches were lack of advanced machine learning methods (e.g., Support Vector Machines), which are useful techniques for data classification and regression, and used only one or a few parameters (e.g., SST, SSH, SSS or dynamic height) to synthesize the subsurface thermal structure. Thus, the estimation methods, as well as the accuracy, still have room for improvement.

Support Vector Machines (SVMs), a popular suite of machine learning models, are very adept at data classification and regression, as well as pattern recognition (Borges, 1998, Huang et al., 2002, Weston and Watkins, 1999). This paper proposes a novel approach based on support vector machines (supervised learning models) to estimate subsurface temperature anomaly (STA) in the Indian Ocean from satellite remote sensing measurements including sea surface temperature anomaly (SSTA), sea surface height anomaly (SSHA), and sea surface salinity anomaly (SSSA) with the help of Argo in situ data for SVM training and testing, all of these sea surface parameters in large scale can now be obtained through satellite remote sensing platforms. The validations were carried out by comparing the horizontal maps and vertical profiles of SVM-estimated STA with Argo STA data at different depth levels.

Section snippets

Study area and data

The Indian Ocean (IO) is the third largest of the world's oceans. The Indian Ocean is the warmest ocean in the world, and is largely influenced by a monsoon climate. The ocean's currents are mainly controlled by the monsoon. In recent years, the Indian Ocean has been discovered to have a much larger impact on climate variability than previously thought. Over the past decades, IO sea surface temperatures and heat content have increased, and model studies suggest that decadal trends play a ...

Support vector machines

Support vector machines (SVMs) are a popular machine learning method for classification, regression, and other learning tasks (Chang and Lin, 2001, Hsu et al., 2010). SVMs are learning machines derived from the structural risk minimization (SRM) principle (Vapnik, 1995). SVMs are supervised learning models with associated learning algorithms that analyze data and recognize patterns, used for classification and regression analysis. The basic idea behind SVM is to map multidimensional data into a ...

Results and discussions

Fig.2 shows the monthly Sea surface temperature anomaly (SSTA), sea surface height anomaly (SSHA) and sea surface salinity anomaly (SSSA) of the Indian Ocean from satellite measurements (the sea surface parameters as input datasets for SVR) with the same $1^{\circ} \times 1^{\circ}$ spatial resolution for October 2010. The SSTA, SSHA and SSSA range from -3 to 3°C , -0.2 to 0.2m and -1.5 to 1.0 PSS in general, respectively (Fig.2). The sea surface parameters show distinctive spatial heterogeneity over the whole ...

Conclusions

In the study, we proposed a support vector machine method that was a new technique for studying the subsurface thermal structure via satellite remote sensing. SVM, one of the most popular machine learning methods, is a useful advanced technique in large scale subsurface and deeper ocean thermal remote sensing. SVM can accurately estimate the STA upper 1000m in the Indian Ocean from satellite measurements of sea surface parameters. The results were reliable with reasonable accuracy by validating ...

Acknowledgments

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Citation Excerpt:

...Previous studies have shown that the deeper ocean remote sensing (DORS) technique, combined with float data, has a great potential to indirectly retrieve the ocean interior from satellite observations (Su et al., 2015; Su et al., 2018a). DORS methods include data assimilation and numerical modeling (Klemas and Yan, 2014), dynamic theoretical approach (Wang et al., 2013; Liu et al., 2019; Yan et al., 2020), and empirical statistical approach (Su et al., 2015; Jeong et al., 2019). Because of the complexity and uncertainty of the numerical and dynamic modeling, their accuracy for large-scale subsurface modeling and estimation is not guaranteed....

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